

HARINE S/O SUMEN

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PROFESSIONAL SUMMARY

Computer Science undergraduate at SIT / DigiPen with an Electronic and Computer Engineering diploma, driven by how software works under the hood. Tech lead on a custom C++ engine for a published game; owned the Go backend and Jenkins CI/CD for an Android vision app. Also a teacher, chess player and composer, seeking software engineering roles.

EDUCATION

Singapore Institute of Technology, in collaboration with DigiPen Institute of Technology Singapore *Sep 2024 – Present*
BSc (Hons) in Computer Science in Real-Time Interactive Simulation *Expected Graduation: Aug 2028*

Ngee Ann Polytechnic *Apr 2019 – May 2022*
Diploma in Electronic and Computer Engineering

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Go, Kotlin, Java, TypeScript, Lua, GDScript, VBA

Engine & Game Development: ECS Architecture, OpenGL, ImGui, Lua Scripting, Memory Management, Input Handling

Backend & Infrastructure: Go + SQLite APIs, Jenkins CI/CD, Linux Servers, Nginx, Git LFS, Gitea, Tailscale

Tools: GitHub, Visual Studio, CMake, Godot, Android Studio, Jira, FMOD, Aseprite, DaVinci Resolve, Microsoft Access VBA

Additional: Software Architecture, Test Automation, Documentation, Calculus, Leadership, Teaching, Music Composition

PROJECTS

Technical Lead | TTBEEngine — Custom C++ Engine for TiltTiltBoom (Published Game) *Sep 2025 – May 2026*

- Designed **TTBEEngine's** modular C++ ECS architecture, ImGui editor and hot-reloaded Lua scripting as top committer, enabling an 8-person team of programmers, designers and artists to iterate in one codebase as scope grew
- Built the controller system (gamepads, two-axis gyro aiming, haptics) and a deterministic replay system for **TiltTiltBoom's** screen-tilting mechanic, delivering precise controls and per-level leaderboards
- Set up CMake targets and an Inno Setup installer and composed the soundtrack, leading to the game's public release

Technical Lead | PinPoint — Android Bowling Coach App *Sep 2026 – Present*

- Built a **Go + SQLite API server** with invite-only crypt-hashed accounts, token sessions, upload limits and rate limiting, securing user accounts and private videos, backed by 11 test files
- Set up **Jenkins CI/CD** that builds signed APKs and deploys behind go vet/go test gates with automatic rollback, and hardened the VM (HTTPS, Tailscale, fail2ban), preventing broken builds from reaching users

Creator | Aaron — Python Webcam Security Guard (Open Source) *Sep 2026*

- Built **Aaron**, a webcam guard using OpenCV face recognition with Telegram control, offline voice wake (Vosk) and MediaPipe gestures, locking Windows automatically when a stranger sits down
- Split the app into three processes via file-based IPC, eliminating a race between Telegram commands and the camera

WORK EXPERIENCE

Teaching Assistant | DigiPen Institute of Technology Singapore (Hybrid) *Jan 2026 – Apr 2026*

- Ran calculus crash courses for 50+ students per session and 1-on-1 consultations, breaking abstract concepts into step-by-step explanations, improving students' understanding

Software Engineer Intern | Denso Wave Incorporated *Oct 2021 – Feb 2022*

- Built VBA-based inventory management systems and Python automation scripts for microcontroller validation, improving workflow efficiency and reducing manual testing
- Debugged, tested and documented processes with cross-functional engineers, which led to improved internal tools

LEADERSHIP & ACTIVITIES

President & Quartermaster | SIT Mindsports *Jan 2025 – Dec 2025*

- Led the club and managed its equipment and logistics while competing in 10 chess competitions, placing **3rd Runner-Up** as a team in Singapore's Intervarsity League against all universities, polytechnics and JCs

FYE Helix Assistant Head Facilitator | Singapore Institute of Technology *May 2025 – Aug 2026*

- Co-led the facilitator team running FYE orientation activities, helping freshmen settle into university life